

Smart Waste Management System for Sustainable Cities

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Abstract - Rapid urbanization and changing consumption patterns are increasing the volume and complexity of municipal solid waste, straining conventional collection and disposal practices, and contributing to bin overflow, illegal dumping, and low recycling rates. This paper proposes an integrated smart waste management framework that combines multimodal AI for disposal guidance, a Pay-As-You-Throw-inspired fee mechanism informed by LSTM based waste prediction and XGBoost based tax regression, a bilingual (English/Sinhala) RAG based conversational assistant over municipal documents, and a temporal LSTM based forecasting framework for ward-level waste demand prediction, integrated with resource allocation and route optimization for efficient municipal waste collection. Experiments on real world inspired datasets show strong waste-classification performance, accurate weekly waste prediction, improved response relevance in the conversational agent, and superior forecasting accuracy relative to a purely temporal LSTM baseline, indicating the potential of the proposed system to support sustainable, data-driven urban waste management.

Keywords- *Smart Waste Management, Multimodal AI, Behavior Based Taxation, Conversational Agents, Municipal Solid Waste, LSTM, XGBoost, Internet of Things, Retrieval-Augmented Generation, Sustainable Urban Management, Waste Demand Forecasting, Predictive Analytics*

I. INTRODUCTION

Municipal solid waste management is a growing challenge in rapidly urbanizing cities, where increasing waste generation, poor segregation, bin overflow, illegal dumping, and inefficient collection practices negatively impact public health, environmental sustainability, and

municipal efficiency. Conventional systems often rely on fixed collection schedules, limited real-time monitoring, and manual decision-making, resulting in inefficient resource utilization and low recycling effectiveness. Additionally, citizens often lack proper disposal guidance, while existing fee structures provide limited incentives for responsible waste management.

To address these challenges, this research proposes an integrated smart waste management system that combines AI-based disposal guidance, IoT-enabled bin monitoring, behaviour based smart taxation, bilingual conversational support, and LSTM-based waste demand forecasting with resource allocation and route optimization. The proposed framework aims to improve waste segregation, reduce overflow, promote sustainable citizen behaviour, and enhance municipal collection efficiency. Scientifically, it contributes by integrating multimodal AI, machine learning, retrieval-augmented generation, and predictive analytics into a unified framework for sustainable urban waste management, supporting data-driven decision-making for both citizens and municipal authorities in smart city environments.

II. RELATED WORKS

Recent smart waste management research has mainly focused on waste classification, IoT enabled bin monitoring, and waste generation forecasting. Computer vision and text-based models have been applied to identify waste categories and support source separation [1]. However, many of these solutions stop at category

prediction and do not provide personalized disposal techniques such as reuse, recycling, or upcycling with practical step-by-step guidance. IoT based waste systems commonly monitor current bin fill levels and support collection planning [2]. However, such systems are mostly reactive because they show the present condition of the bin but do not predict when a bin is likely to overflow. Forecasting based approaches use temporal models to predict bin status or waste generation trends [3], some of existing forecasting approaches are commonly developed as separate modules and are not directly linked with citizen facing disposal support. This research addresses these limitations by combining image/text-based waste classification, personalized disposal recommendations, bin capacity checking, and overflow prediction in a single citizen-facing disposal support module.

Economic approaches such as Pay-As-You-Throw (PAYT) have been introduced to encourage waste reduction by charging users based on volume or weight [4]–[6]. These schemes can reduce residual waste and improve source separation, but most implementations rely on fixed tariff structures or directly observed waste amounts. As a result, they provide limited support for predictive, behaviour aware fee calculation and may not respond well to changing household waste patterns.

Conversational support for waste services is commonly provided through static FAQ systems or rule based chatbots [7]. Recent studies also show growing interest in AI-based public service assistants [8]. However, without proper grounding in official municipal documents, such systems may provide incomplete or incorrect guidance, especially where collection schedules, bin rules, and service procedures vary by location and language.

Temporal forecasting methods, including LSTM based models, have shown value in learning seasonal and short-term patterns in urban service demand [9], [10]. Related smart city studies also highlight the importance of connecting prediction outputs with resource allocation decisions [11]. However, many existing works still treat forecasting as a separate analytical task rather than linking it with demand scoring, crew allocation, and route optimization. Therefore, the main research gap is the lack of an integrated smart waste management framework that connects citizen-level disposal guidance, behaviour-based taxation, bilingual information support, and municipal-level demand forecasting within one operational platform. The novelty of this research is its unified AI, IoT, machine learning, and conversational assistance framework designed to support both citizens and municipal authorities in sustainable waste management.

III. METHODOLOGY

A. System Architecture Overview

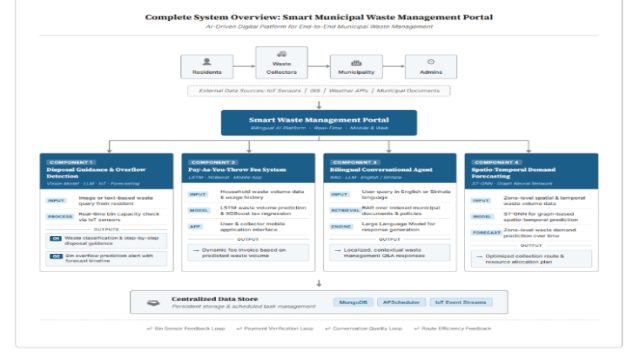


Figure 1: System Architecture Diagram

The proposed system follows a multi-layer architecture that connects data collection, AI models, and user-facing applications. As shown in Figure 1, resident inputs, IoT bin data, and municipal records are processed through four main modules: AI-based disposal guidance, LSTM XGBoost smart fee prediction, bilingual RAG-based citizen support, and LSTM-based ward demand forecasting. These outputs are delivered through a resident mobile app, collector app, and municipal planning dashboard, enabling disposal guidance, bin-fit decisions, tax monitoring, resource allocation, and route optimization within one integrated platform.

B. Integrated System Components

1. End-to-End Disposal Guidance with Capacity Checking and Overflow Forecasting

This component delivers end-to-end disposal guidance by allowing users to submit a waste image or text description, which is classified into the correct waste category. Based on the category and user context, the system recommends the most suitable technique such as reuse, recycle, or upcycle, followed by step-by-step disposal guidance [12]. It also checks whether the selected smart bin has enough remaining capacity using IoT sensor readings converted into fill-level estimates [13], [14]. Finally, a lag-based time-series forecasting model predicts possible future overflow, helping users make correct disposal decisions and supporting municipalities in reducing bin overflow risk [15].

2. Household Waste-Based Smart Tax or Fee Calculation System

This component introduces a transparent behaviour based smart fee model by combining LSTM-based household waste forecasting with XGBoost based tax prediction. LSTM is used to forecast weekly waste quantities and handle temporal waste patterns, while XGBoost estimates tax values from predicted or recorded waste data [16], [17],

[20]. Since fine grained household data may be unavailable, aggregated waste records are converted into realistic household level data using a simulation approach [18]. The final tax is calculated using a deterministic formula to keep the process fair, reproducible, and auditable, while mobile app features such as rankings, rewards, and collector verification encourage responsible waste behaviour [19], [21].

3. Conversational Agent for Waste Issue Resolution and Service Guidance.

This component provides a bilingual RAG-based conversational assistant that gives citizens accurate and localized waste management information from official municipal documents [22]. The system extracts and indexes document content using FAISS for semantic retrieval and BM25 for keyword matching, especially for Sinhala location names and collection schedules. User queries are detected as English or Sinhala, then relevant document segments are retrieved, reranked, and filtered according to the user’s registered location. English responses are generated using LLaMA 3.1 8B, while Sinhala responses use a QLoRA-tuned Gemma model [23]. By grounding answers in official documents, the assistant reduces misinformation, supports questions on collection days, segregation rules, and complaints, and lowers routine inquiry workload for municipal staff [24].

4. Urban Service Demand Forecasting and Resource Optimization Using Temporal Machine Learning

This component supports city-level waste planning by using an LSTM model to predict ward level waste demand from complaint records, service logs, bin clearance data, estimated waste volume, and weather factors. The predicted demand is converted into risk categories and demand scores, helping municipal planners identify wards that need urgent attention. Based on these scores, the system allocates crews and collection resources according to priority and workload. A route optimization module then generates efficient collection routes by considering depot locations, ward priorities, and travel distance, while a web dashboard visualizes risk maps, service queues, and recommended resource deployment for better data-driven municipal planning.

IV. RESULTS AND DISCUSSION

The integrated system was evaluated module-wise to assess the feasibility and effectiveness of the proposed architecture. Overall, the results indicate robust performance across system functions. In the disposal

guidance pipeline, the EfficientNetB0 based image classifier showed the training accuracy behavior of the waste classification model, indicating that the model learns category-level waste features effectively during training in Figure 2.



Figure 2: Model Training Accuracy Graph

The text-based classifier also performed well on descriptive queries, with most errors arising from generic or ambiguous inputs. In addition, the XGBoost-based technique recommendation model demonstrated reliable prediction of whether reuse, recycling, or upcycling was most suitable for a given item and user profile. Finally, scenario-based evaluations that coupled classification with IoT derived bin status suggested fewer simulated overflow events than a non-adaptive baseline. The key model performances are summarized in Table 1, highlighting strong results for text classification, technique prediction, and overflow prediction within the disposal guidance component.

Task/Model	Metric	Value
Text classification (API model)	Accuracy	0.98
Technique prediction (XGBoost)	Accuracy	0.85
Overflow prediction (Random Forest)	Accuracy	8.37%

Table 1 Models Metrics Values

In the Smart tax calculation system, the LSTM models accurately predicted weekly household waste. They showed low RMSE and MAE along with high R^2 values. These results captured seasonal patterns and household behaviors, consistent with earlier LSTM based forecasting studies [16], [17]. The XGBoost based tax model matched reference values well and performed well on outliers, supporting the use of gradient boosting for environmental tabular datasets [20]. Overall, the hybrid AI and deterministic framework delivered realistic billing results while keeping a clear link between penalties or discounts and user behavior.

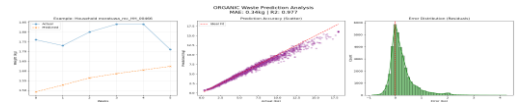


Figure 3: Model Waste Weight Prediction Analysis in Organic

The comparison shown in Figure 3 demonstrates that predicted waste weights closely follow the actual values across the three waste categories used in the smart tax

calculation process. Simulations over several months showed that high-waste households consistently faced higher fees. In contrast, low waste households enjoyed discounts and recognition on the leaderboard. This pattern matches previous gamification research, which suggests that incentives and social comparison can encourage lasting behavior change. [21]. The regression results in Table 2 confirm that the XGBoost tax prediction model produces highly accurate fee estimates, with a strong R^2 value and low prediction errors.

Model	Regression Metrics	Value
Tax prediction (XGBoost)	R^2	0.9989
	RMSE	Rs. 3.68
	MAE	Rs. 0.78
	MAPE	1.85%

Table 2 Tax Prediction Model Metrics Values

For the conversational assistant, cross encoder re-ranking improved the relevance of retrieved document segments. A hybrid retrieval strategy that combined keyword based (BM25) and dense semantic search performed well for Sinhala queries that included location names and collection schedules, aligning with observations that hybrid retrieval methods outperform single-strategy baselines in RAG systems [22]. Human evaluation indicated that responses were accurate and aligned with municipal documents when relevant context was retrieved. Location based filtering improved precision for schedule-related queries by restricting candidates to the user's registered area. Overall, grounding responses in official documents reduced the risk of hallucinated rules, and bilingual support improved accessibility compared to the available waste chatbots [7].

Metric	English Mean	Sinhala Mean
BLEU	0.635	0.513
ROUGEL	0.774	0.450
BERTScore-F1	0.709	0.964

Table 3 English & Sinhala RAG evaluation

The English RAG agent in Table 3 achieves higher lexical and sequence overlap (BLEU, ROUGE-L values), while maintaining reasonable semantic similarity (BERTScore-F1), reflecting good performance on English queries with occasional variations in phrasing. The Sinhala RAG agent demonstrates moderate lexical and sequence overlap with reference answers while achieving high semantic fidelity, indicating accurate understanding and generation of Sinhala responses. The Sinhala evaluation in Table 3 shows high semantic similarity, suggesting that the assistant preserves the intended meaning even when the exact word overlap is lower.

For city-scale forecasting, the proposed system applies a temporal LSTM based prediction model to capture historical waste generation patterns within each municipal ward. By learning seasonal trends, short-term demand fluctuations, and environmental influences from historical operational data, the model improves the identification of high-risk wards that require immediate attention in municipal collection planning. Event driven case analyses further indicated that the model effectively captures sudden increases in service demand caused by factors such as complaint surges, missed collections, or adverse weather conditions, enabling more reliable detection of operational hotspots. These forecasts are integrated into a municipal planning dashboard that supports exploration of alternative resource allocation and route planning scenarios through an intuitive interface for decision-makers.

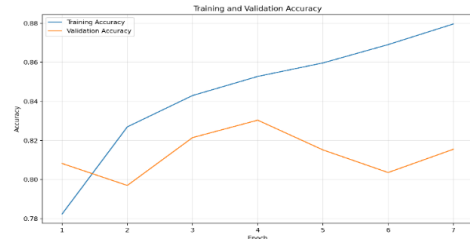


Figure 4: Model Training Accuracy Graph

The accuracy pattern shown in Figure 4 suggests that the LSTM model learns temporal waste generation patterns that can support ward-level forecasting and municipal resource planning.

Task/Model	Metric	Value
Waste demand prediction (LSTM)	Accuracy	0.781
	Precision (Marco Avg)	0.781
	Recall (Marco Avg)	0.747
	F1-Score (Marco Avg)	0.759
	Best Validation (Marco Avg)	0.830

Table 4 LSTM Waste Demand Prediction Metrics

The LSTM demand prediction results in Table 4 indicate that the model can classify ward-level waste demand risk with acceptable accuracy, precision, recall, and F1-score values.

V. CONCLUSION

This paper presented an integrated smart waste management framework that combines multimodal AI, IoT-enabled bin monitoring, behaviour based fee prediction, bilingual RAG assistance, and LSTM-based demand forecasting within a unified platform. The system supports waste classification, overflow prediction, smart fee calculation, localized citizen guidance, resource allocation, and route optimization, bridging citizen-level waste management with municipal planning [12]-[24].

The results demonstrate the framework's ability to improve waste identification, reduce overflow risk, support transparent fee estimation, enhance multilingual citizen services, and enable data-driven waste collection planning. Overall, the proposed approach provides a practical and sustainable foundation for efficient, citizen-entered smart waste management in urban environments [13]-[24].

VI. FUTURE WORK

Future work will focus on piloting the system in partner municipalities and collecting longitudinal, anonymized operational data to calibrate models further and evaluate fairness impacts. Additional improvements include expanding localized training data to enable more robust classification, strengthening the conversational assistant through improved document updating and citation, and linking demand forecasting to route optimization and traffic-aware scheduling. Establishing standardized interfaces and governance mechanisms will also be important for responsible integration with other smart-city services.

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